PYTHON FOUNDATIONS - ASSIGNMENT 03

Assignment & Conditional Logic

15 beginner-friendly scenario challenges: 5 on assignment operators and 10 on comparison operators, logical conditions, if, if-else and elif.

`=` `+=` `-=` `*=` `/=` `==` `!=` `>` `<`

`if / else / elif`

Name: _____

Date: _____

Batch: _____

BEGINNER TRACK

15 USE-CASE CHALLENGES

Quick reference before you begin

Part A - Assignment operators: update an existing variable. **Part B - Conditions:** comparisons produce `True` or `False`, and `if` / `else` uses that result to decide which code should run.

= Assign value	**+=** Add and assign	**-=** Subtract and assign	***=** Multiply and assign
/= Divide and assign	**==** Equal to?	**!=** Not equal to?	**> < >= <=** Compare values

1 **`=` vs `==`**

`=` stores a value. `==` asks whether two values are equal.

2 **Conditions return Boolean values**

A comparison such as `age >= 18` becomes `True` or `False`.

3 **`if` runs conditionally**

The indented block runs only when its condition is true.

4 **`elif` checks another condition**

Use it when there are more than two possible outcomes.

Logical operators: `and` requires both conditions, `or` requires at least one, and `not` reverses a Boolean result.

Assignment rule: Use only the concepts covered so far. No loops, lists, dictionaries or custom functions are required.

Assignment Operators

Update the same variable step by step instead of creating a new variable after every operation.

1 Digital Wallet Balance Tracker

`=` `+=` `-=`

A user starts with a wallet balance, receives money, spends money and gets a cashback.

1. Take the opening wallet balance from the user.
2. Take a salary credit and add it using `+=`.
3. Take shopping expense and subtract it using `-=`.
4. Take cashback received and add it using `+=`.
5. Take electricity bill payment and subtract it using `-=`.
6. Print the balance after every update and the final balance.

2 Game Score Updater

`+=` `-=` `*=`

A player earns points, loses points and then receives a multiplier bonus.

1. Take the player's starting score.
2. Add points won in Level 1 using `+=`.
3. Subtract penalty points using `-=`.
4. Add Level 2 points using `+=`.
5. Take a score multiplier such as 2 and update the score using `*=`.
6. Print the score after each stage and the final score.

3 Warehouse Stock Update

`+=` `-=` `//=`

A warehouse tracks stock as items arrive, are sold, and are packed into equal cartons.

1. Take the opening stock quantity.
2. Add newly delivered items using `+=`.
3. Subtract sold items using `-=`.
4. Subtract damaged items using `-=`.
5. Take items per carton and use a separate variable with `//=` to calculate complete cartons.
6. Display updated stock and number of complete cartons.

4 Monthly Savings Progress

`+=` `-=` `/=`

A student tracks savings after deposits, purchases and an equal sharing calculation.

1. Take the current savings amount.
2. Add pocket money using `+=`.
3. Add gift money using `+=`.
4. Subtract the cost of a book using `-=`.
5. Copy the final savings into another variable and use `/=` to calculate the amount if shared equally among 3 siblings.
6. Print final savings and equal share.

5 Mobile Data Balance Simulator

`+=` `-=` `*=` `/=`

A user starts with mobile data, buys an add-on, consumes data and then evaluates a promotional multiplier.

1. Take the starting data balance in GB.
2. Add a data add-on using `+=`.
3. Subtract today's usage using `-=`.
4. Subtract yesterday's pending usage using `-=`.
5. Copy the current balance to a promotional variable and double it using `*=`.
6. Copy the balance to another variable and use `/=` to find an equal daily allowance for 5 days.
7. Print original remaining data, promotional data and daily allowance.

Challenge: Explain in one sentence why `x += 5` and `x = x + 5` produce the same updated value.

Checkpoint: Before moving on, students should be able to read `balance -= 500` as: "take the current balance, subtract 500, and store the new value back into balance."

Conditions with if / else

Use comparison operators to produce True or False, then let Python decide which block should run.

6 Driving Age Eligibility

`>=` `IF-ELSE`

A simple system checks whether a person is old enough for a driving-licence application.

1. Take the person's name and age.
2. If age is 18 or above, print "Eligible to apply".
3. Otherwise print "Not eligible yet".
4. Also calculate how many years remain until age 18 and display it only for the not-eligible case.

7 Pass or Fail Result

`>=` `IF-ELSE` `AVERAGE`

A student enters marks in three subjects and the program decides the result using the average.

1. Take marks for Maths, Science and English.
2. Calculate total and average.
3. If average is 40 or above, print "PASS".
4. Otherwise print "FAIL".
5. Print total and average in both cases.

8 Even or Odd Number Checker

`%` `==` `IF-ELSE`

Use the modulo operator with a condition to decide whether a number is even or odd.

1. Take an integer from the user.
2. Find the remainder when divided by 2.
3. If the remainder is equal to 0, print "Even Number".
4. Otherwise print "Odd Number".
5. Also display the remainder.

9 Positive, Negative or Zero

`>` `<` `==` `IF-ELIF-ELSE`

A number can fall into exactly one of three categories.

1. Take a number from the user.
2. If the number is greater than 0, print "Positive".
3. Else if it is less than 0, print "Negative".
4. Otherwise print "Zero".
5. Print the number together with its category.

10 Largest of Two Numbers

`>` `==` `IF-ELIF-ELSE`

Compare two values and handle the special case where both are equal.

1. Take two numbers from the user.
2. If the first is greater, print the first as the larger number.
3. Else if the second is greater, print the second as the larger number.
4. Otherwise print "Both numbers are equal".

11 Student Grade Calculator

`>=` `ELIF` `RANGES`

A school assigns a grade based on percentage.

1. Take percentage from the user.
2. 90 or above: Grade A.
3. 75 to below 90: Grade B.
4. 60 to below 75: Grade C.
5. 40 to below 60: Grade D.
6. Below 40: Fail.
7. Print the percentage and final grade.

Think carefully about order: check the highest range first, then move downward.

12 Movie Ticket Category

`<` `>=` `ELIF`

A cinema sets ticket prices based on age category.

1. Take customer name and age.
2. Age below 5: ticket is free.
3. Age 5 to 12: ticket price is 150.
4. Age 13 to 59: ticket price is 300.
5. Age 60 or above: ticket price is 180.
6. Print customer name, category and ticket price.

13 Simple Login Checker

`==` `and` `IF-ELSE`

A training application has one fixed username and password.

1. Take username and password from the user.
2. Assume the correct username is `student` and password is `python123`.
3. If both username and password are correct, print "Login Successful".
4. Otherwise print "Invalid Credentials".
5. Use the logical operator `and` to combine the two comparisons.

14 Shopping Discount Eligibility

`>=` `and / or` `IF-ELSE`

A store gives a discount when a customer meets its offer rules.

1. Take purchase amount and membership status as `yes` or `no`.
2. Give a discount if purchase amount is at least 5000 AND the customer is a member.
3. Print "Discount Applied" or "No Discount".
4. Then modify the condition so a discount is allowed when purchase amount is at least 8000 OR the customer is a member.
5. Run both conditions and compare the results.

15 Employee Performance Bonus - Master Challenge

`>=` `and` `ELIF` `NESTED THINKING`

A company decides bonus level using performance score and attendance.

1. Take employee name, performance score out of 100 and attendance percentage.
2. If score is at least 90 AND attendance is at least 90, print "Excellent - 20% Bonus".
3. Else if score is at least 75 AND attendance is at least 80, print "Good - 10% Bonus".
4. Else if score is at least 60 AND attendance is at least 75, print "Satisfactory - 5% Bonus".
5. Otherwise print "No Bonus".
6. Take monthly salary and calculate the bonus amount using the applicable percentage.
7. Print employee name, category, bonus amount and salary after bonus.

Master concept: This question combines arithmetic, comparison operators, logical operators, variables, type conversion and conditional statements.

Before you submit

Self-check: Your goal is not just to get output. You should be able to explain why each operator and each condition is being used.

- I know the difference between `=` and `==`.
- I can rewrite `x = x + 5` as `x += 5`.
- I understand that comparisons return `True` or `False`.
- I use indentation correctly after `if`, `elif` and `else`.
- I know when to use `if` only.
- I know when to use `if-else`.
- I know when multiple ranges need `elif`.
- I can use `and` when both conditions must be true.
- I can use `or` when either condition can be true.
- I test boundary values such as 18, 40, 60, 75 and 90.

Next recommended topic: After this assignment, teach nested `if` in more depth, then `for` loops.